

Twenty years ago youtube published the first video «me at the zoo» 

1. Suppose that salaries $x_1, x_2, \dots, x_n$ (measured in units of the minimum wage) are independent and follow a simple Pareto distribution with probability density function

$$f(x | \alpha) = \frac{\alpha}{x^{\alpha+1}}, \quad x \geq 1,$$

where $\alpha > 1$ is an unknown shape parameter.

- (a) [4] Find the method of moments estimator $\hat{\alpha}_{\text{MM}}$.
 (b) [5] Let $p \in (0, 1]$ be the proportion of the population with the highest salaries. Show that the share of total income received by this top p fraction of the population is

$$D(p) = p^{\frac{\alpha-1}{\alpha}}.$$

- (c) [1] The empirical Pareto principle (the «80/20 rule») states that approximately 20% of the population receives 80% of the total income. Using the formula find $\hat{\alpha}_{\text{rule}}$ that satisfies

$$D(0.2) = 0.8.$$

2. Let $x_1, x_2, \dots, x_n$ be a random sample from a binomial distribution $\text{Bin}(N, p)$ with probability mass function

$$P(x_i = x) = C_N^x p^x (1-p)^{N-x}, \quad x = 0, 1, 2, \dots, N.$$

- (a) [4] Find the method of moments estimators for N and p .

Consider known $N = 4$ and the observed sample $(4, 1, 2, 2, 1)$.

- (b) [3] Find the MLE estimate of p .
 (c) [2] Prove that MLE estimator of p is unbiased and consistent.
 (d) [1] Find a minimal sufficient statistics for p .
 3. The depth of corrosion μ [mm] on a metal plate is measured using four different unbiased independent methods (ultrasonic, electrical resistance, optical, and X-ray). Due to the different principles and equipment, the measurement variances are known to be

$$\sigma_1^2 = 1.6, \quad \sigma_2^2 = 2, \quad \sigma_3^2 = 2.5, \quad \sigma_4^2 = 3.$$

The measured values [mm] of obtained by the four methods are

$$x_1 = 19, \quad x_2 = 18, \quad x_3 = 20, \quad x_4 = 21.$$

- (a) [3] Give at least one example of an unbiased estimator $\tilde{\mu}$ of μ .
 (b) [4] Find the linear unbiased estimator $\hat{\mu}$ of μ that has the smallest possible variance.
 (c) [1] Compute the numerical value of $\hat{\mu}$ using the given data.
 (d) [2] Compute the standard error of $\hat{\mu}$.

4. Random variables $X_1, \dots, X_n$ are independent with Poisson distribution $\mathbb{P}(X_i = k) = \exp(-\lambda)\lambda^k/k!$ for non-negative integers k . Random variables $Y_1, \dots, Y_n$ are independent with $\mathbb{P}(Y_i = 1) = p$ and $\mathbb{P}(Y_i = 0) = 1 - p$.

I do not observe X_i nor Y_i , I observe only their product $W_i = X_i Y_i$. Among $n = 1000$ observations of W_i : 300 observations are exactly equal to 0 and $\sum W_i = 7000$.

Test the null hypothesis $p = 1$ against alternative $p < 1$ using likelihood ratio test with significance level 0.05.

FYI: right 0.05 critical chi-squared values are $\chi_{1,\text{crit}}^2 = 3.84$, $\chi_{1,\text{crit}}^2 = 5.99$, $\chi_{1,\text{crit}}^2 = 7.81$.

5. Random variables $X_1, X_2, \dots, X_n$ are independent and identically distributed with

$$\mathbb{P}(X_i = 0) = \exp(-a), \quad \mathbb{P}(X_i = k) = \mathbb{P}(X_i = -k) = \exp(-a)a^k/(2 \cdot k!) \text{ for } k \in \mathbb{N}.$$

Let $Y_i = |X_i|$.

- (a) [6] Calculate Fisher information for the sample $X_1, \dots, X_n$ and for the sample $Y_1, \dots, Y_n$. Which sample contains more information about unknown a ?
- (b) [4] Does maximum likelihood estimator based on $X_1, \dots, X_n$ attain the Cramer-Rao boundary?

6. Random variables $X_1, X_2, \dots, X_n$ are independent and identically distributed with

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- (a) [4] Find minimal sufficient statistic T for the unknown parameter a .

Let $g(X_1, X_2, \dots, X_n)$ be some function that does not depend on a directly.

- (b) [3] Is it possible that $\mathbb{E}(T \mid g(X_1, X_2, \dots, X_n))$ depends on a ?
- (c) [3] Is it possible that $\mathbb{E}(g(X_1, X_2, \dots, X_n) \mid T)$ depends on a ?